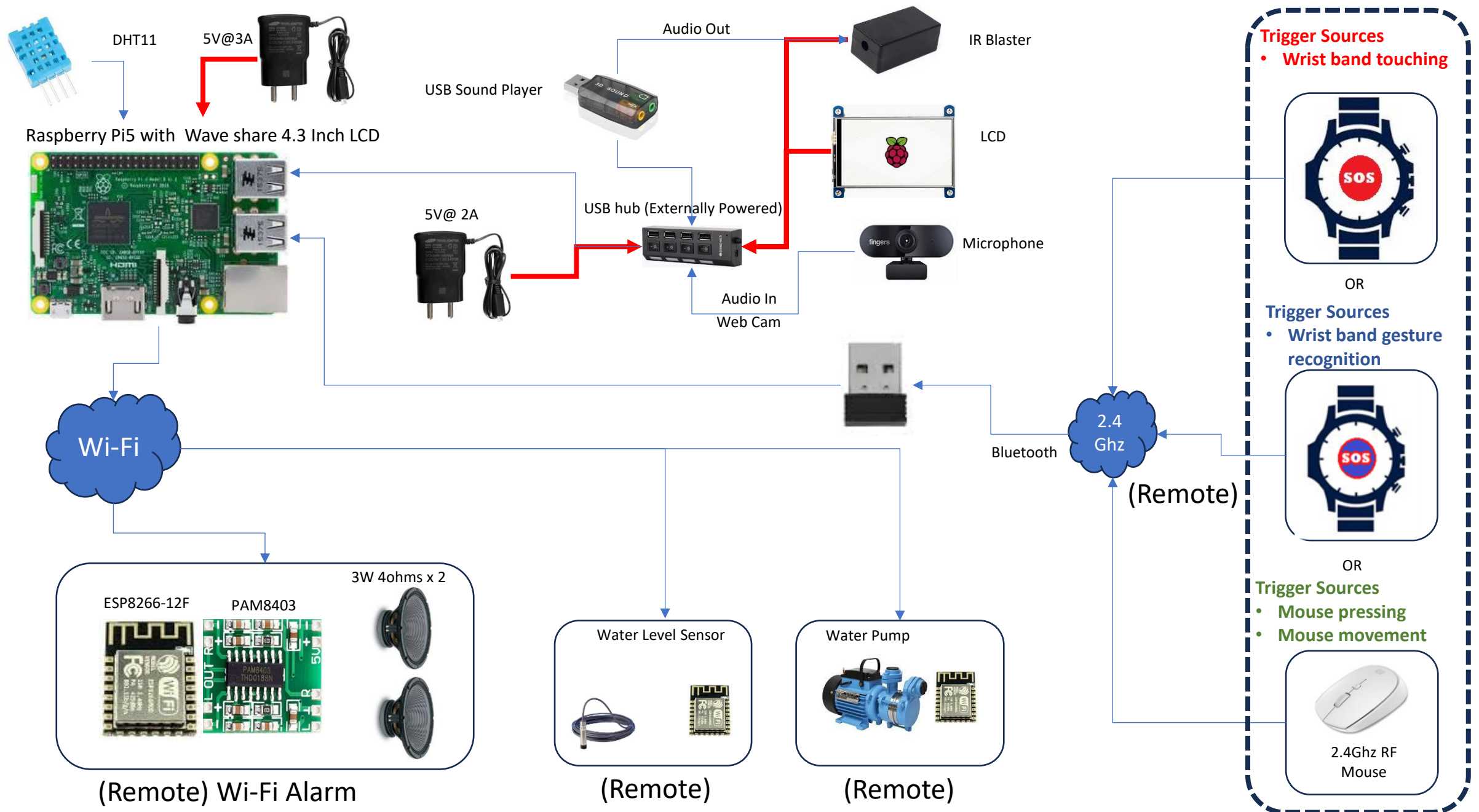


Use Case Wi-Fi Alarm Speaker + [Touch or Gesture or Mouse] Alarm Source + Solar-Grid-Water Level Monitoring





Voice
Activation



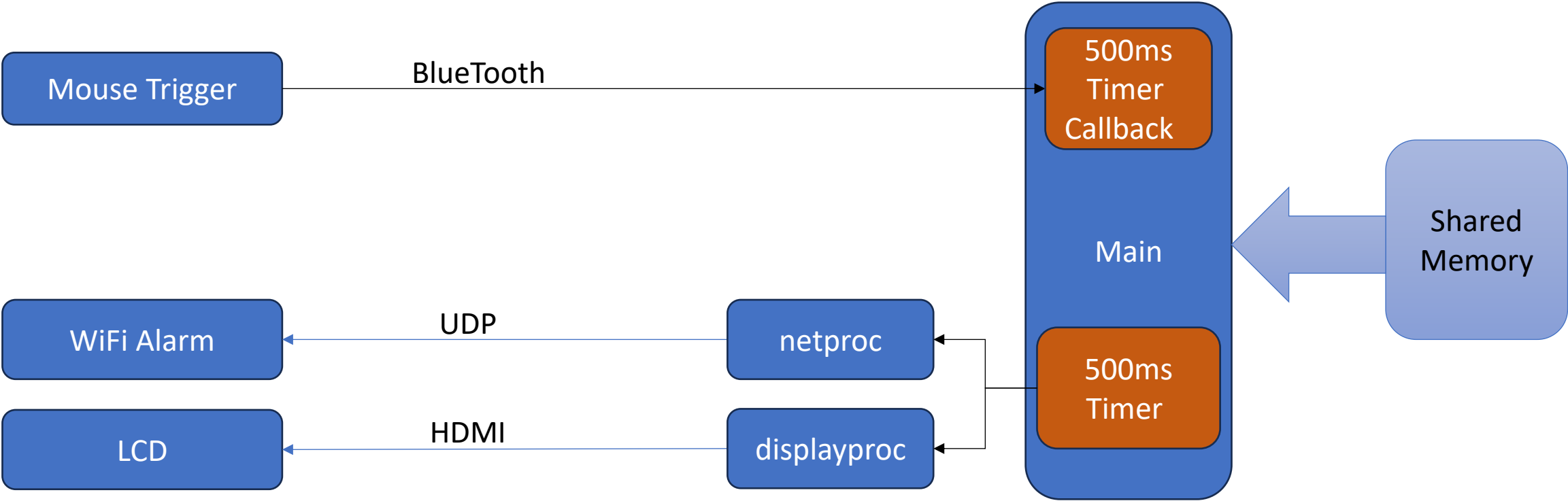
Wrist Band
Activation



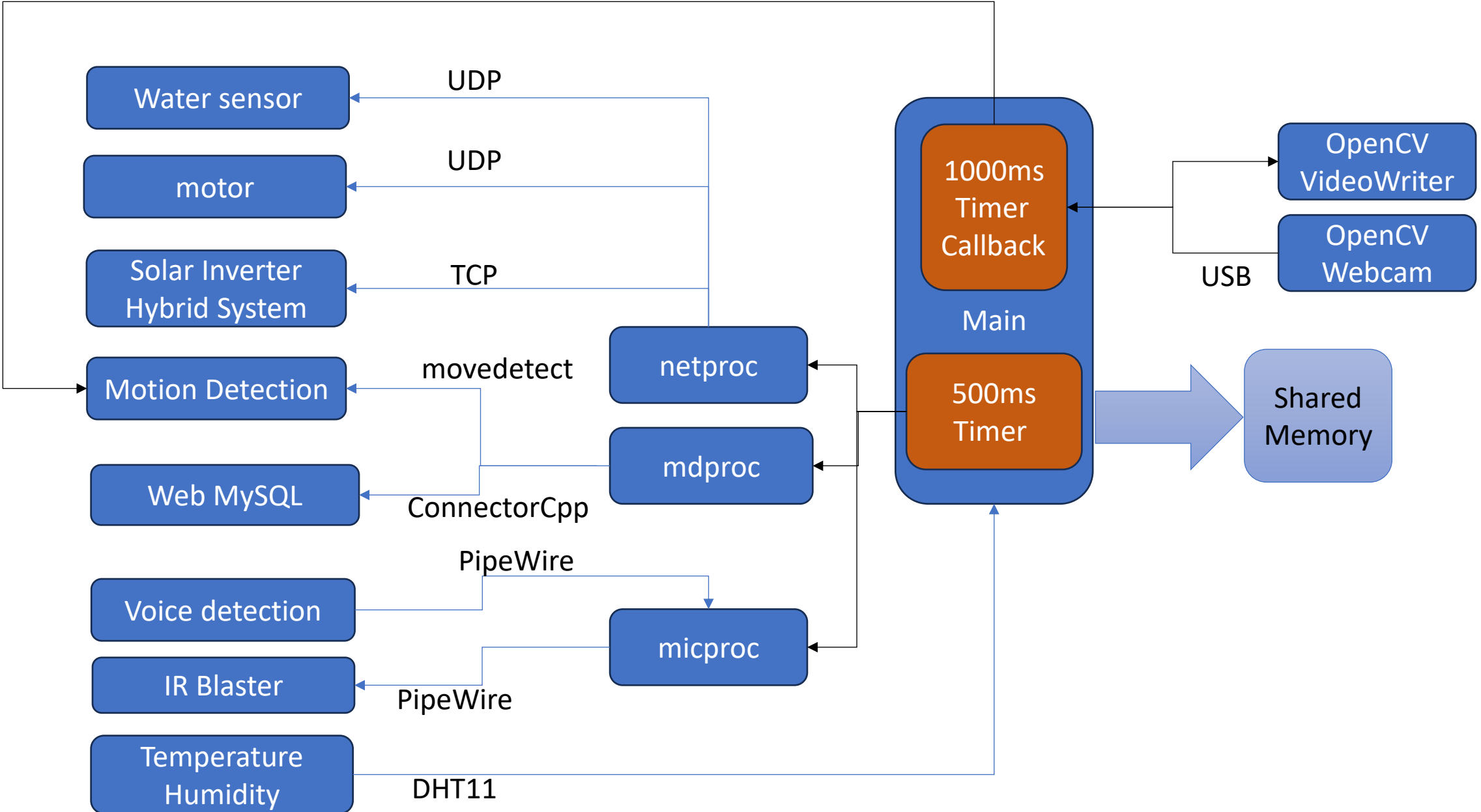
Motion Detection
Activation

ALARM

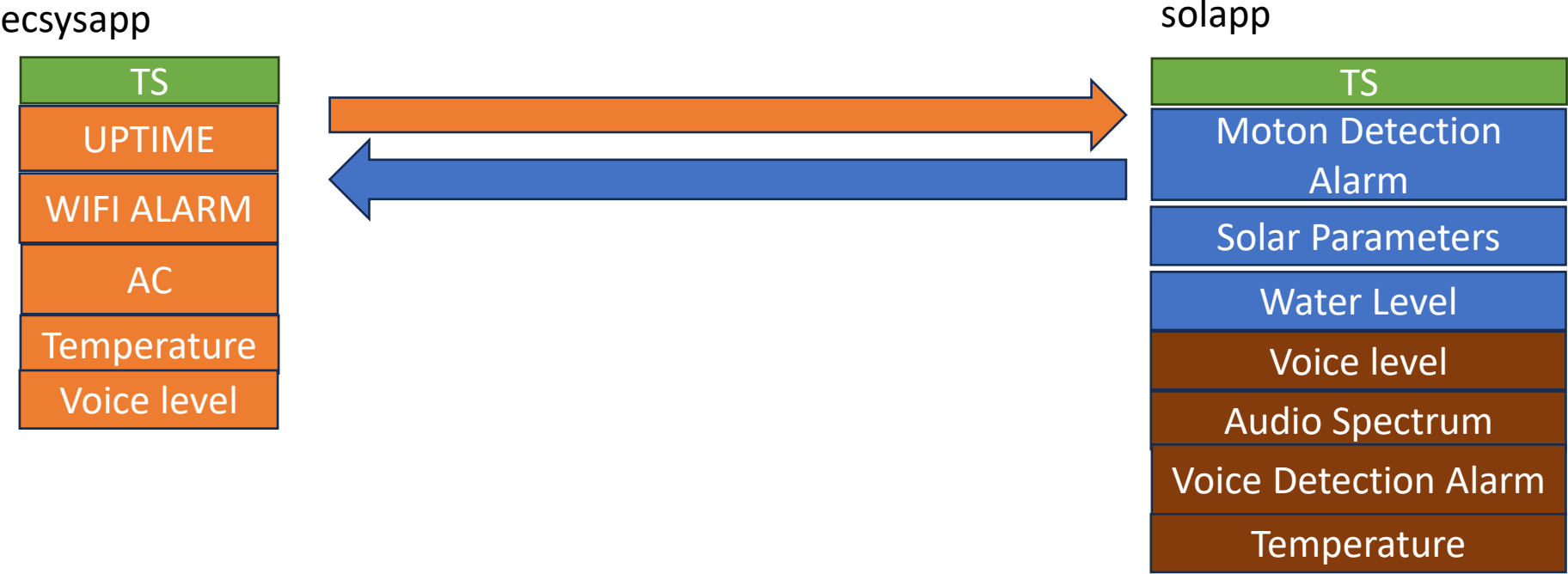




solapp



Shared Memory Data

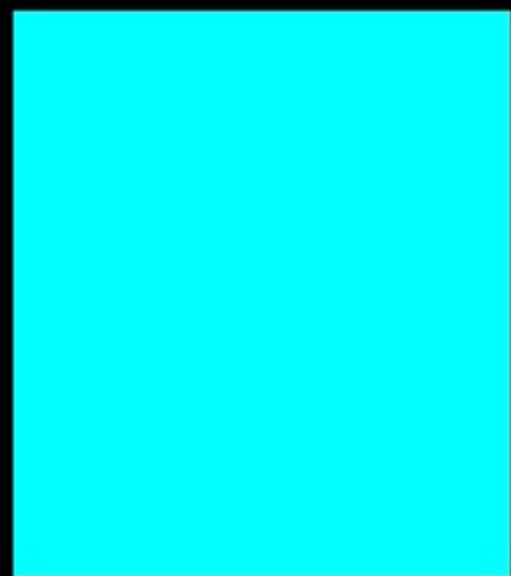


WL[82%]

BAT[38%]

SPWR[38%]

GRID_PWR



11:00:27

UPTIME [0:0:7]



27 c

LOAD 277W

System Configurations

ID	NAME
0	2.4G Mouse
1	2.4G Mouse
2	C-Media Electronics Inc. USB PnP Sound Device
3	pwr_button
4	Lenovo FHD Webcam Audio: Lenovo
5	vc4-hdmi-0
6	vc4-hdmi-0 HDMI Jack
7	vc4-hdmi-1
8	vc4-hdmi-1 HDMI Jack

Application Parameters

MOUSE_LEVEL_A[0-5]	0	
MOUSE_LEVEL_B[0-5]	0	
MOUSE_LEVEL_B_HOURS[0,1,2...23]	DEFAULT	
MOUSE_NAME	2.4G Mouse	
MOUSE_INDEX	1	
BEACON_TIMEOUT[1-255]	15	
DIR_MAX[1-255]	2	
NIGHT MODE HOURS[0,1,2...23]	20,21,22,23,0,1,2,3,4,5	
AC ON HOURS[0,1,2...23]	11,13,14,20,22	
ALARM KEY		
ALARM IP	192.168.0.105	
AC CUT OFF TEMPERATURE	26	
VOICE DETECTION HOURS[0,1,2...23]	12,13,14,15	
MOTION DETECTION HOURS[0,1,2...23]	DEFAULT	
SOLAR IP	192.168.0.100	
MOTOR IP	192.168.0.106	
SENSOR IP	192.168.0.107	
Solar Serial Number		
MOTOR KEY		
SENSOR KEY		
Motor Hour	12	
Water Low	220	
Water High	270	
ACCESS_PASSWORD	admin	Submit

Select the image area for motion detection

ID 22 X 0 Y 0 W 320 H 480

Mask AddMask Remove

X

Y

W

H

Add

ID

Remove

160

320

480

640

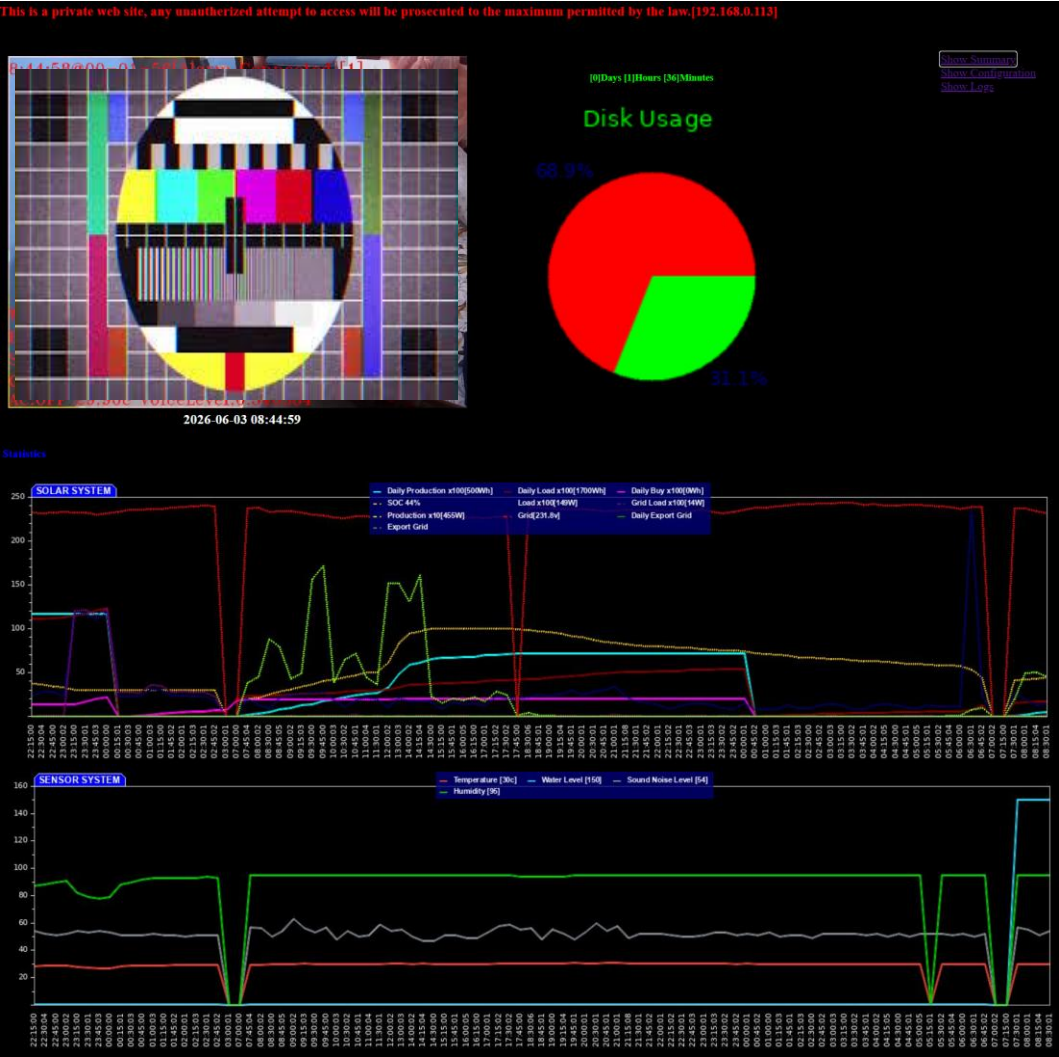
120

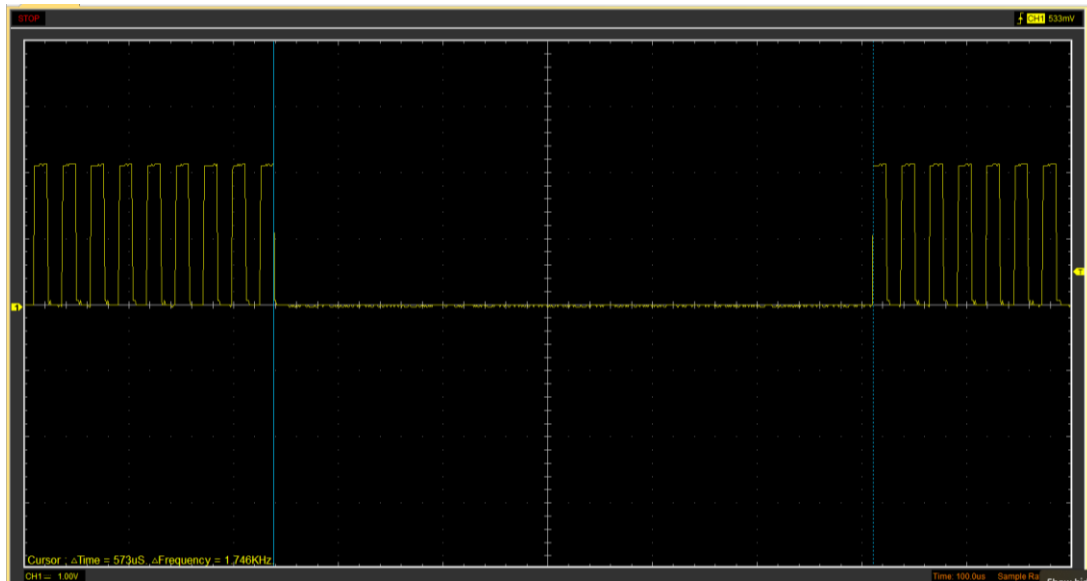
240

360

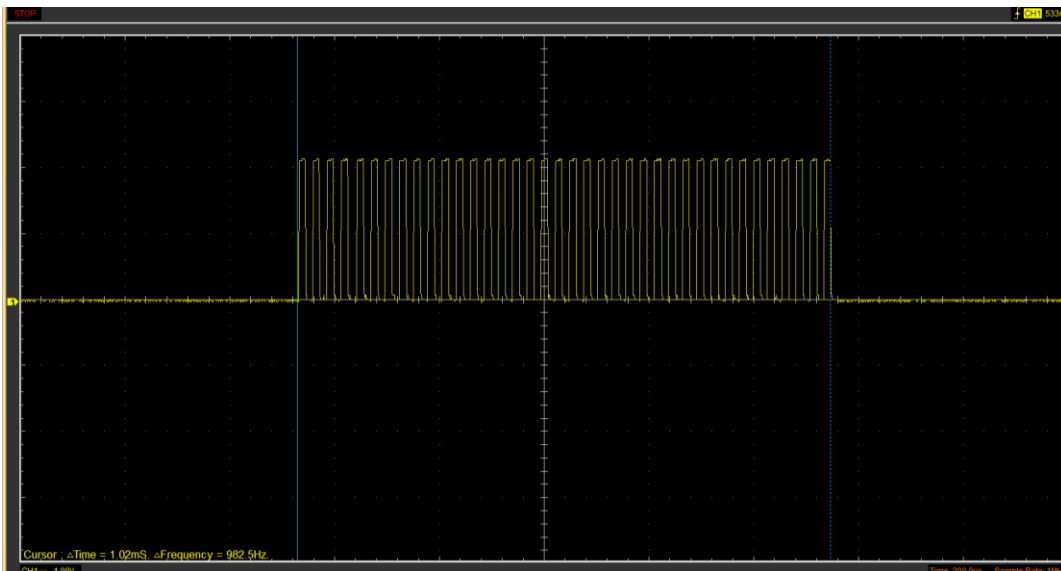
480

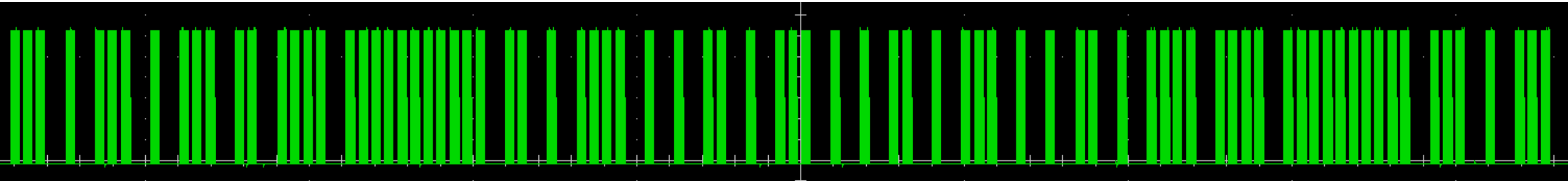
REBOOT





VOLTAS AC

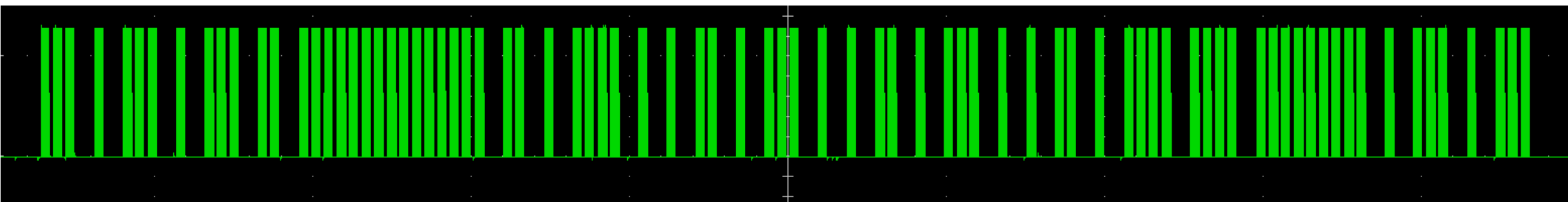




1010100000100000101010000010000010101000001010000010101010000010101010101010101010100000101000001000
0010101010000010000010000010100000100000101010000010000010000010100000100000101010000010000010000010
100000100000101010100000101010100000101010101010101010000010101000001000001010100000

ON

2A082A082A0A0AA0AAAAA82820AA0820A082A0820A082A082A082A82A82AAAA82A082A0

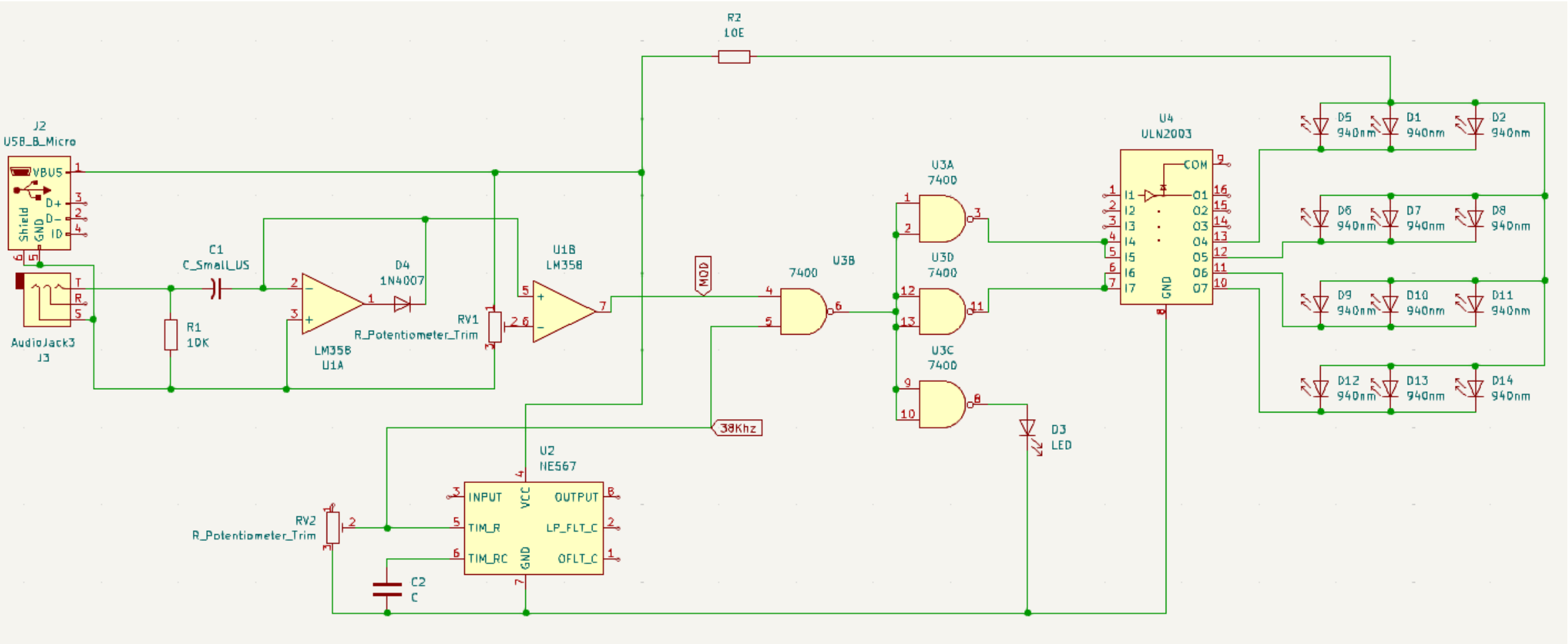


10101000001000001010100000100000101010000010100000101010101010101010101010101010101000001010000010000010
1010100000100000100000101000001000001010100000100000100000101000001000001010100000100000100000101000
001000001010101000001010101000001010101010101010100000100000101010000010000010101

OFF

A820A820A8282AAAAAAA0A082A82082820A82082820A82082841541541555504150415

IR Blaster Schematics



.bashrc

```
cmd=$(pgrep keep_alive.sh)
if [ -z "$cmd" ]; then
    if [ -f /home/ecsys/enable ]; then
        /home/ecsys/ecsysapp/keep_alive.sh &
    fi
fi
```

cmdline.txt

```
console=serial0,115200 console=tty1 root=PARTUUID=b64a6f0e-02
rootfstype=ext4 fsck.repair=yes rootwait ds=nocloud;i=rpi-imager-
1778722754770 cfg80211.ieee80211_regdom=IN logo.nologo
fbcon=rotate:3 vc4.force_hotplug=1 usbcore.autosuspend=-1
```

config.txt

```
[all]
gpio=26=op,dh
dtoverlay=dht11,gpiopin=21
```

- sudo nmcli con add con-name ethernet ifname eth0 type ethernet ip4 10.10.10.1/24 gw4 10.10.10.1
- sudo nmcli con mod ethernet ipv4.dns 10.10.10.1,8.8.8.8
- sudo nmcli con mod ethernet connection.autoconnect true
- sudo nmcli con up ethernet ifname eth0

- sudo nmcli connection add type 802-11-wireless con-name **ssid_name** ssid **ssid_name** 802-11-wireless-security.key-mgmt WPA-PSK 802-11-wireless-security.psk **password**
- sudo nmcli con mod **ssid_name** connection.autoconnect true
- sudo nmcli connection up **ssid_name**

/var/www/html

- sudo chown www-data:www-data -R *
- sudo setfacl -m u:ecsys:rwX storage

```
sudo apt-get install mariadb-server php cmake libmysqlcppconn-dev syslog-ng acl vim universal-ctags apache2 libboost-all-dev libopencv-dev  
https://github.com/stephanecharette/MoveDetect  
sudo apt install php8.4-mysql php8.4-gd libfmt-dev ttf-mscorefonts-installer libdrm-dev libpipewire-0.3-dev libfftw3-dev
```

```
create table syscon(id MEDIUMINT,name CHAR(255) NOT NULL);  
create table mask(id int NOT NULL AUTO_INCREMENT,x MEDIUMINT NULL,y MEDIUMINT NULL,w MEDIUMINT NULL,h MEDIUMINT NULL,PRIMARY KEY  
(id));  
create table cfg(mouse_levela TINYINT NULL,mouse_levelb TINYINT,mouse_bhrs CHAR(64), mouse_name CHAR(255) NOT NULL,mouse_index MEDIUMINT  
NULL,beacon_timeout MEDIUMINT NULL,dir_max MEDIUMINT NULL,night CHAR(64) NULL,ac CHAR(64) NULL,akey CHAR(64) NULL,aip CHAR(32) NULL,aco  
MEDIUMINT NULL,voice CHAR(64) NULL,motion CHAR(64) NULL,sip CHAR(32) NULL,mip CHAR(32) NULL,wip CHAR(32) NULL,sn CHAR(32) NOT NULL,mkey  
CHAR(32) NOT NULL,wkey CHAR(32) NOT NULL,whr MEDIUMINT NULL,wlo MEDIUMINT,whi MEDIUMINT,access CHAR(255) NOT NULL,reboot TINYINT  
NULL);
```

```
create table hour(temp smallint NULL,noise smallint NULL,wl smallint NULL,dprod smallint NULL,dload smallint NULL,dbuy smallint NULL,soc smallint  
NULL,upload smallint NULL,gload smallint NULL,prod smallint NULL,gvolt smallint NULL,gdexp smallint NULL,gexp smallint NULL,ts TIMESTAMP NOT NULL  
DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP);
```

```
create table out_img(data mediumblob,ts TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP);  
insert into out_img(data) values(LOAD_FILE('/home/seccam/backup/dbase/tv640.jpg'));
```

```
create user "userecsys"@"localhost"identified by "ecsys123";  
grant all on ecsys.* TO "userecsys"@"localhost";
```

```
sudo nmcli dev wifi rescan  
nmcli dev wifi list  
sudo nmcli device wifi rescan && nmcli device wifi list
```

```
sudo systemctl --no-pager status syslog-ng  
sudo systemctl restart syslog-ng  
sudo systemctl enable syslog-ng
```

```
MariaDB [ecsys]> desc cfg;
```

Field	Type	Null	Key	Default	Extra
mouse_levela	tinyint(4)	YES		NULL	
mouse_levelb	tinyint(4)	YES		NULL	
mouse_bhrrs	char(64)	YES		NULL	
mouse_name	char(255)	NO		NULL	
mouse_index	mediumint(9)	YES		NULL	
beacon_timeout	mediumint(9)	YES		NULL	
dir_max	mediumint(9)	YES		NULL	
night	char(64)	YES		NULL	
ac	char(64)	YES		NULL	
akey	char(64)	YES		NULL	
aip	char(32)	YES		NULL	
aco	mediumint(9)	YES		NULL	
voice	char(64)	YES		NULL	
motion	char(64)	YES		NULL	
sip	char(32)	YES		NULL	
mip	char(32)	YES		NULL	
wip	char(32)	YES		NULL	
sn	char(32)	NO		NULL	
mkey	char(32)	NO		NULL	
wkey	char(32)	NO		NULL	
whr	mediumint(9)	YES		NULL	
wlo	mediumint(9)	YES		NULL	
whi	mediumint(9)	YES		NULL	
access	char(255)	NO		NULL	
reboot	tinyint(4)	YES		NULL	

25 rows in set (0.001 sec)

```
MariaDB [ecsys]> desc out_img;
```

Field	Type	Null	Key	Default	Extra
data	mediumblob	YES		NULL	
ts	timestamp	NO		current_timestamp()	on update current_timestamp()

2 rows in set (0.003 sec)

```
MariaDB [ecsys]> desc mask;
```

Field	Type	Null	Key	Default	Extra
id	int(11)	NO	PRI	NULL	auto_increment
x	mediumint(9)	YES		NULL	
y	mediumint(9)	YES		NULL	
w	mediumint(9)	YES		NULL	
h	mediumint(9)	YES		NULL	

5 rows in set (0.001 sec)

```
MariaDB [ecsys]> desc hour;
```

Field	Type	Null	Key	Default	Extra
temp	smallint(6)	YES		NULL	
humd	smallint(6)	YES		NULL	
noise	smallint(6)	YES		NULL	
wl	smallint(6)	YES		NULL	
dprod	smallint(6)	YES		NULL	
dload	smallint(6)	YES		NULL	
dbuy	smallint(6)	YES		NULL	
soc	smallint(6)	YES		NULL	
upload	smallint(6)	YES		NULL	
gload	smallint(6)	YES		NULL	
prod	smallint(6)	YES		NULL	
gvolt	smallint(6)	YES		NULL	
gdexp	smallint(6)	YES		NULL	
gexp	smallint(6)	YES		NULL	
ts	timestamp	NO		current_timestamp()	on update current_timestamp()

15 rows in set (0.001 sec)

```
Database changed
MariaDB [ecsys]> show tables;
```

Tables_in_ecsys
cfg
hour
mask
out_img
syscon

5 rows in set (0.002 sec)

```
MariaDB [ecsys]> desc syscon;
```

Field	Type	Null	Key	Default	Extra
id	mediumint(9)	YES		NULL	
name	char(255)	NO		NULL	

2 rows in set (0.003 sec)